THE UNIVERSITY OF AZAD JAMMU AND KASHMIR

Department of Software Engineering | Muzaffarabad

ACADEMIC LAB REPORT

LAB 01

Software Construction & Development

UML Use Case Diagrams

SEMESTER	**Fall 2026**
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REGISTRATION NO.	2024-SE-11

SCOPE

Q1: Online Ticket Reservation System | Q2: Credit Card Processing System

INTRODUCTION + QUESTION 1

Online Ticket Reservation System

Objective and theory

The objective of this lab is to identify system users (actors) and their interactions with a system using **UML use case diagrams**. A use case diagram gives a high-level view of functional requirements.

In UML, **stick figures** represent external actors, **ovals** represent use cases, the **rectangle** defines the system boundary, and an **«include»** relationship shows functionality reused as part of another use case.

Q1 | Use case model

Scenario: A customer can search available tickets, make a reservation, complete payment, and obtain an electronic confirmation. The administrator maintains ticket details and availability.

Actors: Customer, Administrator, and Payment Service. **Core use cases:** Search Tickets, Check Availability, Reserve Ticket, Process Payment, Issue E-Ticket, and Manage Tickets/Schedules.

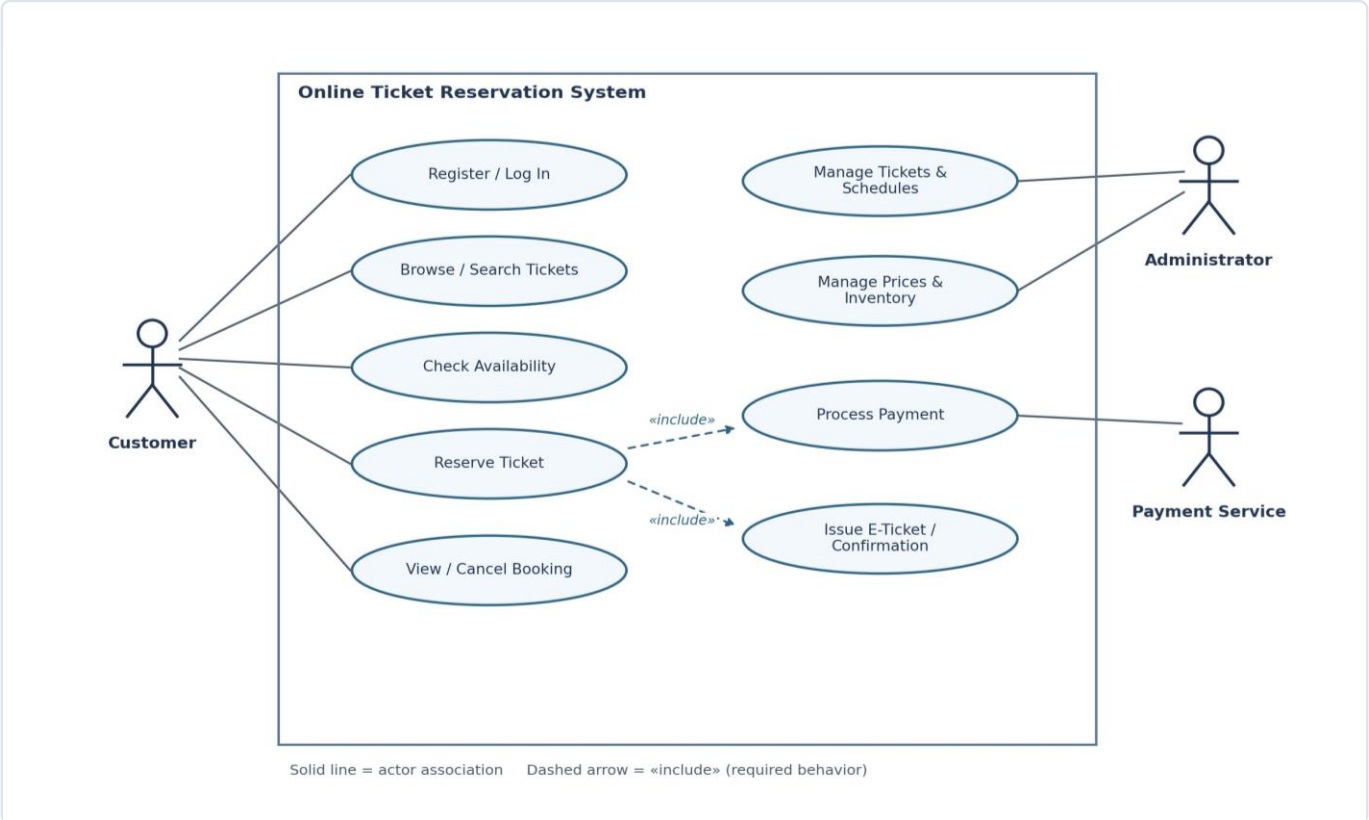


FIGURE 1 UML use case diagram for the Online Ticket Reservation System.

Interpretation

The customer initiates booking-related actions. Reserving a ticket includes payment processing and e-ticket confirmation as modeled; the Payment Service interacts with the payment use case.

QUESTION 2 | CASE STUDY

Credit Card Processing System

Problem statement: A customer selects items in a shop. The vendor prepares the bill, reads the customer's credit card, and submits the payment amount. The system checks/updates the available credit and prints a receipt or balance statement; the customer then signs the receipt.

Actors and responsibilities

- **Customer:** purchases goods using a credit card and signs the printed receipt.
- **Vendor / Cashier:** prepares the bill, swipes/reads the card, enters the amount and provides the receipt.
- **Bank / Card Issuer:** an external supporting actor that authorizes payment and verifies credit limits.

Use cases: Record Selected Items & Prepare Bill; Pay by Credit Card; Swipe/Read Card; Enter Transaction Amount; Authorize Transaction; Check Available Credit; Update Credit Balance; Print Receipt/Balance Statement.

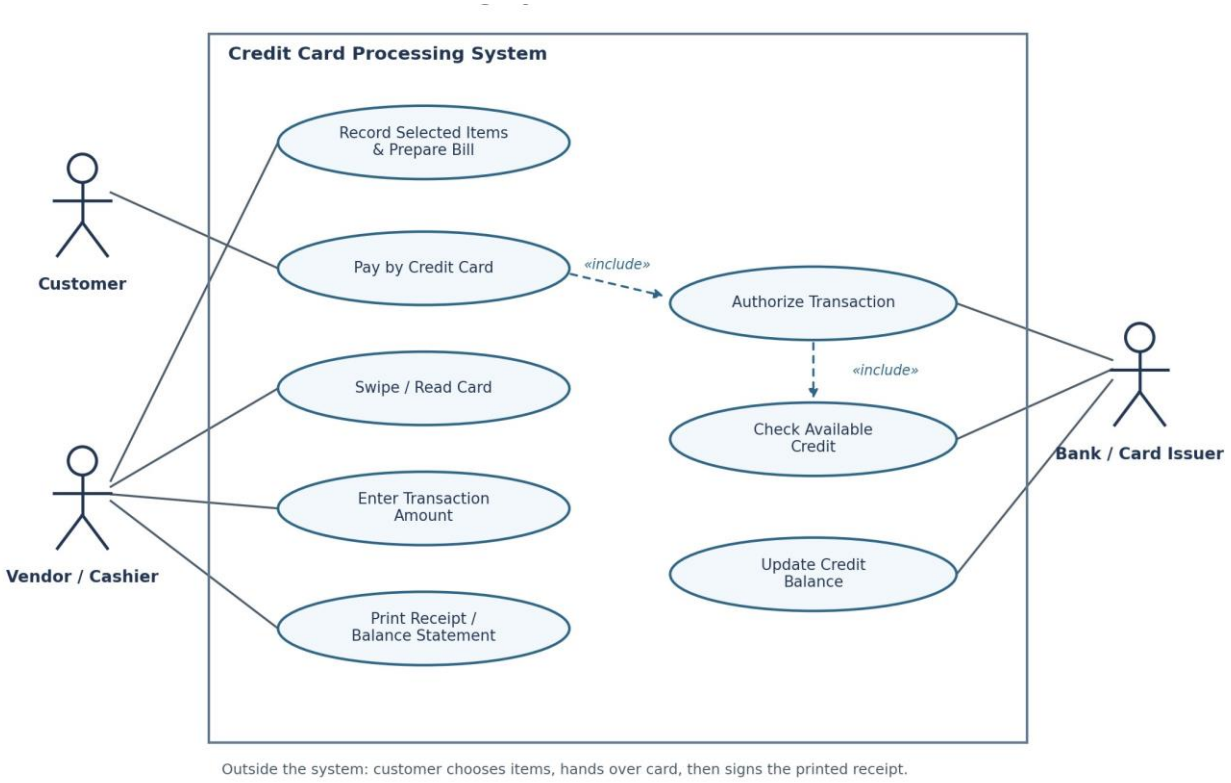


FIGURE 2 UML use case diagram for the Credit Card Processing System.

Conclusion

This lab demonstrates how UML use case diagrams describe a system from the users' viewpoint. The ticket reservation model identifies booking services, while the credit card model identifies payment functions and external authorization. Together they clarify actors, system boundaries, and functional requirements before implementation.